

# Power Architecture Derivations

Relative Institutional Packages and Type-Dependent Participation

Working formal appendix

2026-05-11

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## 1 Status

This is the clean working surface for the reimplemented model architecture. The prior feasibility-branch derivation has been archived in git tag `redesign-feasibility-branch-2026-05-11`. Do not import theorem statements, notation, or branch labels from that archived architecture without rederiving them here.

The active manuscript `formal_model_v5.Rmd` remains frozen. New formal work should be developed in this file first, checked by scripts, independently verified, and only then transported to the paper.

## 1.1 Architecture Decision

The main baseline no longer uses fixed transfers that may fit one realized pie but not another. Proposals are relative institutional packages: quota rules, production shares, cut obligations, enforcement provisions, exceptions, and flexibility terms. These packages are feasible in every state.

Screening comes from H's type-dependent participation threshold:

$$U_H(y, \theta) = y + b_H(\theta),$$

where  $y$  is the concession offered by weak states and  $b_H(\theta)$  is the direct benefit that H obtains from being inside the agreement. H accepts if the package payoff exceeds its continuation value.

## 2 Q&A

### What is the substantive puzzle?

When can unanimity or consensus favor a powerful hegemon even though the rule appears to constrain its formal power? In the current baseline, this is a conditional institutional-payoff question, not an endogenous rule-choice stage.

### What mechanism should the model isolate?

Unanimity can make an informed hegemon pivotal. Weak states must offer a package that satisfies H's participation constraint, but they do not observe how favorable the package must be for H to prefer agreement to its outside option.

### What is the key modeling reset?

The proposal is a relative institutional package, not an absolute transfer. It is always feasible. The private-information problem is about H's type-specific acceptance threshold.

### What is the OPEC interpretation?

Members bargain over quotas, cuts, shares, monitoring, exceptions, and flexibility. They can observe the proposed package, but they do not know Saudi Arabia's true participation threshold because Saudi Arabia privately knows its spare capacity, opportunity cost, and outside option.

## 3 Follow-up Q&A for This Proof Pass

**Does the clean baseline use weak-state recognition in both rounds?** Yes. For this proof pass,  $\pi_H = 0$  in both rounds. H is pivotal under unanimity but is not a formal agenda setter.

**What happens after terminal Round-2 rejection?** The game ends. H receives an external disagreement payoff  $d_H(\theta)$ , and weak states receive their normalized disagreement payoff zero.

**Is  $y$  a fixed transfer from the institutional pie?** No.  $y$  is a relative institutional concession. It is always feasible as a package, but it reduces the residual payoff available to the weak coalition one-for-one.

**What is being checked here?** This pass resets the primitives for a fixed-pie relative-package model. All previous Round-2 formulas are treated as superseded until they are rederived from these primitives.

## 4 Primitives

There are  $N \geq 3$  states: one hegemon  $H$  and  $m = N - 1$  weak states. Nature draws

$$\theta \in \{0, 1\}.$$

The hegemon observes  $\theta$ . Weak states share prior belief

$$\mu = \Pr(\theta = 1).$$

The institutional surplus available to the weak coalition is fixed and known. Normalize it to

$$B = 1.$$

The private state  $\theta$  affects  $H$ 's participation threshold, not the size of the weak coalition's pie.

The discount factor is  $\beta \in (0, 1]$ .

Agenda power is represented by

$$\pi_H \in [0, 1].$$

The baseline sets

$$\pi_H = 0,$$

so only weak states are recognized as proposers. This isolates informational power through pivotality from proposal power.

## 5 Relative Package

A proposal is a package

$$x \in X.$$

For the first reimplementation, use a scalar concession index

$$y \in [0, \bar{y}],$$

where a higher  $y$  is more favorable to  $H$  and less favorable to the weak coalition. The package is feasible if  $y \in [0, \bar{y}]$ . Feasibility does not depend on  $\theta$ .

The fixed-pie baseline restricts

$$0 \leq \bar{y} \leq 1.$$

Substantively,  $y$  may represent a larger production share for H, a weaker cut obligation, more flexibility, better enforcement against other members, or more favorable treatment under changing market conditions.

## 5.1 H Payoff

The minimal payoff representation is

$$U_H(y, \theta) = y + b_H(\theta). \quad (\text{H-payoff})$$

The term  $b_H(\theta)$  is the direct benefit H obtains from the agreement itself, apart from the concession  $y$ . It may capture cartel coordination, price stability, predictability, reputation, or the value of constraining other members.

Let

$$C_H(\theta, \mu')$$

be H's continuation value after rejection, given the posterior  $\mu'$ . H accepts package  $y$  in state  $\theta$  iff

$$y + b_H(\theta) \geq \beta C_H(\theta, \mu'). \quad (\text{H-PC})$$

Define the type-specific acceptance threshold

$$y_\theta^*(\mu') = \beta C_H(\theta, \mu') - b_H(\theta). \quad (\text{threshold})$$

The screening condition is

$$y_1^*(\mu') > y_0^*(\mu'). \quad (\text{screening})$$

Equivalently, in a one-shot participation version with outside option  $o_H(\theta)$ ,

$$y_\theta^* = o_H(\theta) - b_H(\theta),$$

so screening requires

$$o_H(1) - b_H(1) > o_H(0) - b_H(0).$$

The high type requires a larger concession because its outside option net of the direct benefit of agreement is higher.

## 5.2 Weak-State Payoffs

We now specify how  $y$  reduces the payoff available to weak states. The normalization is one-for-one in reduced form: increasing the concession to H by one unit reduces the surplus available to the weak coalition by one unit.

The maintained reduced-form payoff primitive is:

$$\text{weak coalition surplus after accepted package } y = 1 - y. \quad (\text{W-cost})$$

Thus  $y$  is costly one-for-one to the weak coalition. This is not a state-contingent feasibility constraint. The package  $y \in [0, \bar{y}]$  is available in every state; if a package leaves a low residual for weak states, that is a payoff consequence, not physical infeasibility.

When weak state  $i$  is recognized and needs weak voters to approve, each non-proposing weak voter  $j \neq i$  receives its continuation value  $c_W$ . The recognized weak proposer keeps the residual:

$$u_i(y, \theta; c_W) = 1 - y - (m - 1)c_W, \quad u_j(y, \theta; c_W) = c_W. \quad (\text{W-residual})$$

In terminal Round 2, weak states' disagreement payoff is normalized to zero, so  $c_W = 0$ . Hence the recognized weak proposer's terminal payoff from an accepted package is  $1 - y$ . A representative weak state's ex ante Round-2 value is the recognized-proposer value divided by  $m$ , because each weak state is recognized with probability  $1/m$  when  $\pi_H = 0$ .

**Status: adopted primitive for the current relative-package proof pass.**

## 6 Baseline Timing: $\pi_H = 0$

The first proof pass keeps  $\pi_H = 0$  in both rounds.

1. Nature draws  $\theta$ ; H observes  $\theta$ , weak states observe only  $\mu$ .
2. A weak state is recognized as proposer.
3. The proposer offers a relative institutional package  $y$ .
4. Under unanimity, H must accept the package for agreement to form.
5. Weak voters receive continuation-value offers required for coalition approval.
6. If agreement fails, the game proceeds to the next bargaining round with beliefs updated by Bayes' rule on path.

Voting follows the standard Baron–Ferejohn convention at pivotal acceptance constraints: a player whose vote would make agreement pass and who is indifferent between accepting the agreement and receiving its continuation or outside option accepts the agreement. Thus H accepts at its participation threshold when it is pivotal, and weak voters accept offers equal to their continuation values when their vote is pivotal. Non-pivotal tie-breaking is specified explicitly in each voting assessment.

If a weak proposer is indifferent between two payoff-maximizing proposals, the baseline assumes that it selects the proposal that minimizes H's expected payoff under the current belief. This tie-breaking rule is conservative with respect to the paper's main claim that private information allows the hegemon to extract informational rents. A later appendix should compare the baseline with the opposite efficient-agreement tie-breaking rule.

No branch in this baseline relies on a package being feasible in one state and infeasible in another.

## 7 Objects To Derive

### 7.1 R2 Unanimity

Status: fixed-pie derivation.

This subsection derives terminal Round 2 from the fixed-pie primitives above. Let

$$d_\theta = d_H(\theta), \quad b_\theta = b_H(\theta).$$

In terminal Round 2, rejection ends the game. H accepts package  $y$  in state  $\theta$  iff

$$y + b_\theta \geq d_\theta.$$

Define the terminal participation threshold

$$\tau_\theta = d_\theta - b_\theta.$$

The fixed-pie screening domain is

$$0 \leq \tau_0 < \tau_1 \leq \bar{y} \leq 1.$$

The proposer never benefits from choosing a package strictly inside an acceptance region, because weak surplus is  $1 - y$ . Therefore the only accepted packages that can be optimal are:

1. low-only:  $y = \tau_0$ , accepted by low H and rejected by high H;
2. pooling:  $y = \tau_1$ , accepted by both types.

The recognized weak proposer's payoff from low-only is

$$L_2(\mu) = (1 - \mu)(1 - \tau_0).$$

The recognized weak proposer's payoff from pooling is

$$P_2(\mu) = 1 - \tau_1.$$

The weak proposer chooses low-only iff

$$(1 - \mu)(1 - \tau_0) \geq 1 - \tau_1.$$

Solving this inequality gives the derived cutoff

$$\mu_2^* = \frac{\tau_1 - \tau_0}{1 - \tau_0}.$$

This is not a primitive. It is the belief at which the weak proposer is indifferent between the two accepted packages. Because  $0 \leq \tau_0 < \tau_1 \leq 1$ ,  $\mu_2^* \in (0, 1]$ , with an interior cutoff when  $\tau_1 < 1$ .

Under the conservative proposer tie-breaking rule, the equilibrium Round-2 proposal is

$$y_2^*(\mu) = \begin{cases} \tau_0, & \mu \leq \mu_2^*, \\ \tau_1, & \mu > \mu_2^*. \end{cases}$$

Thus the recognized weak proposer's Round-2 value is

$$P_{2,prop}^U(\mu) = \begin{cases} (1 - \mu)(1 - \tau_0), & \mu \leq \mu_2^*, \\ 1 - \tau_1, & \mu > \mu_2^*. \end{cases}$$

The representative weak state's Round-2 continuation value is

$$W_2^U(\mu) = \frac{P_{2,prop}^U(\mu)}{m}.$$

The high type's Round-2 continuation value is

$$C_{H,2}^U(1, \mu) = d_1$$

for all  $\mu$ : it either rejects the low-only offer and receives  $d_1$ , or accepts the pooling offer at its threshold and receives  $d_1$ . The low type's Round-2 continuation value is

$$C_{H,2}^U(0, \mu) = \begin{cases} d_0, & \mu \leq \mu_2^*, \\ d_0 + \tau_1 - \tau_0, & \mu > \mu_2^*. \end{cases}$$

Status: proved for terminal Round 2 under the fixed-pie primitives, the normalized threshold domain, the Baron–Ferejohn weak-acceptance convention, and the conservative proposer tie-breaking rule. The derivation is checked in `scripts/verify_relative_package_R2_piH0.R`.

## 7.2 Voting Protocol for R1 Comparisons

Use the same simultaneous public voting protocol under unanimity and majority. After a proposal is made, the proposer is counted as voting yes. All other states vote simultaneously. Individual votes are publicly revealed after the ballot closes. Equivalently, voting is a secret ballot with each player's vote opened and recorded at the end.

A proposal passes if the number of yes votes meets the institutional quota:

$$Q_U = N$$

under unanimity and

$$Q_M = q = \lfloor N/2 \rfloor + 1$$

under majority. Players accept in indifference.

The public history after the ballot includes the proposal, the full vote vector, and the outcome. If the game continues, beliefs are updated from that public history.

The informational distinction comes from pivotality, not from whether a ballot is recorded. Under unanimity, H is pivotal because every state must vote yes. Under majority, a weak proposer may be able to assemble  $q$  weak-state yes votes without H; H may still cast a public no vote, but that vote is not pivotal on the no-screening majority path.

Therefore the majority comparison is not built by suppressing H's vote. It is built by showing when a weak proposer has no reason to buy H's acceptance. This is not automatic. Under majority, buying H's yes vote can replace one weak voter's yes vote. The no-screening majority benchmark therefore requires a condition under which H is not a cheaper coalition partner than a weak voter.

Status: protocol convention adopted for the next R1 and majority derivations. The detailed assessment is recorded in `quality_reports/2026-05-11_common_voting_protocol_unanimity_majority.md`.

### 7.3 Weak-Vote-Passive Assessment

The R1 unanimity result uses a pure-strategy PBE evaluated under a maintained weak-vote-passive assessment. This is not a refinement of standard PBE, not a D1 or intuitive-criterion restriction, and not a claim about all PBEs under arbitrary off-path beliefs. It is a maintained interpretation of the public voting protocol: weak states do not observe H's type, so their unilateral vote deviations are not treated as direct signals about that type.

The maintained assessment has three parts.

1. **Passive beliefs after weak-state deviations.** Weak states do not observe H's type. Therefore, a unilateral deviation by a weak voter does not by itself change beliefs about H. When H's vote is pooling, the posterior after a weak-vote deviation remains the prior. When H's vote separates types, the posterior is determined by H's vote, not by the weak voter's deviation.
2. **H-vote posterior consistency.** If H's prescribed votes separate the two types, public histories with the low-type vote have posterior zero and public histories with the high-type vote have posterior one. If H's prescribed vote pools the two types, that vote carries no type information.
3. **Weak acceptance at pivotal constraints.** A pivotal voter accepts when the proposed payoff equals its continuation value. Non-pivotal votes are not used to create additional information about H unless H's own prescribed vote is type-dependent.

The substantive justification is the standard no-signaling-what-you-do-not-know logic: weak states have no private information about H's type, so their voting deviations should not be interpreted as signals about that type. The assessment rules out off-path belief assignments in which an uninformed weak voter's unexpected no vote changes beliefs about H; it does not purport to eliminate all such belief systems under unrestricted PBE.

Bayesian updating statements below should be read literally for  $\mu \in (0, 1)$ . At endpoint beliefs, the same payoff formulas are interpreted as degenerate-belief or limiting cases, not as updates after positive-probability histories of both types.



## 7.4 R1 Unanimity

Status: proved for the weak-vote-passive pure-threshold candidates below under the fixed-pie relative-package primitives, simultaneous public voting, passive beliefs after weak-state deviations, and weak acceptance at pivotal constraints. The algebra and candidate comparisons are checked in `scripts/verify_relative_package_R1_piH0.R`. This is not a characterization of all standard PBEs under arbitrary off-path beliefs.

### 7.4.1 R1 primitives

Round 1 uses the same players and type space as Round 2. There are  $m = N - 1$  weak states and one privately informed hegemon  $H$ . Nature draws  $\theta \in \{0, 1\}$ ,  $H$  observes  $\theta$ , and weak states share the prior  $\mu = \Pr(\theta = 1)$ .

The baseline recognition protocol sets  $\pi_H = 0$ . A weak state  $i$  is recognized as proposer. The identity of the proposer is public. The proposer is counted as voting yes.

A Round-1 proposal is

$$s = (y, z_{-i}),$$

where  $y \in [0, \bar{y}]$  is the relative institutional concession to  $H$ , and  $z_j \geq 0$  is the payoff promised to each non-proposing weak voter  $j \neq i$ . The proposal is feasible iff

$$y + \sum_{j \neq i} z_j \leq 1. \quad (\text{R1-feasibility})$$

This is a fixed weak-surplus constraint, not a state-contingent pie constraint. The package  $y$  is available in every state.

After the proposal,  $H$  and all non-proposing weak states vote simultaneously. The proposal passes under unanimity iff every state votes yes. If it passes,

$$u_H = y + b_\theta, \quad u_j = z_j, \quad u_i = 1 - y - \sum_{j \neq i} z_j.$$

If it fails, the game proceeds to terminal Round 2. The public history is the proposal and the full vote vector. Let  $\nu$  denote the posterior probability of  $\theta = 1$  after that public history. Weak states receive  $\beta W_2^U(\nu)$ , and type  $\theta$  of  $H$  receives  $\beta C_{H,2}^U(\theta, \nu)$ .

From the R2 derivation, define the recognized weak proposer's terminal value

$$p_2(\nu) = \max\{(1 - \nu)(1 - \tau_0), 1 - \tau_1\},$$

with the conservative low-only tie-breaking rule at equality. Then

$$W_2^U(\nu) = \frac{p_2(\nu)}{m}, \quad c(\nu) = \beta W_2^U(\nu) \quad (\text{weak-cont})$$

is the Round-1 public continuation value of a weak state after a failed proposal with posterior  $\nu$ . Let

$$D_0(\nu) = C_{H,2}^U(0, \nu) = \begin{cases} d_0, & \nu \leq \mu_2^*, \\ d_0 + \tau_1 - \tau_0, & \nu > \mu_2^*, \end{cases} \quad D_1(\nu) = C_{H,2}^U(1, \nu) = d_1. \quad (\text{H-R2-values})$$

Thus  $\mathbf{H}$ 's Round-1 rejection payoff after posterior  $\nu$  is  $\beta D_\theta(\nu)$ . Define the high type's Round-1 threshold and the low type's high-posterior threshold:

$$a_1 = \beta d_1 - b_1, \quad a_0(1) = \beta D_0(1) - b_0. \quad (\text{R1-thresholds})$$

The argument in  $a_0(1)$  is the posterior used to price the low type's continuation value after a rejected low-only proposal. It is not an exponent.

The weak R1 threshold-order domain for the current screening characterization is

$$0 \leq a_0(1) \leq a_1 \leq \bar{y} \leq 1. \quad (\text{R1-threshold-order-domain})$$

Strict low-only separation requires  $a_0(1) < a_1$ . If  $a_0(1) = a_1$ , low-only separation is blocked but pooling at  $a_1$  remains admissible. If  $a_0(1) > a_1$ , the high-threshold pooling proposal below is not admissible under the stated posterior after an off-path  $\mathbf{H}$  no vote; that reversed-order case is outside the current screening characterization and remains blocked for this proof pass.

#### 7.4.2 Exhaustion of pure threshold candidates

For any fixed posterior  $\nu$ ,  $\mathbf{H}$  accepts a package  $y$  iff

$$y + b_\theta \geq \beta D_\theta(\nu).$$

Under the R1 separation domain, the high type's threshold is above the low type's relevant threshold. Therefore a pure threshold proposal can induce only three acceptance sets for  $\mathbf{H}$ :

1. both types accept;
2. only the low type accepts;
3. neither type's acceptance is needed because the proposal is deliberately rejected and the game continues.

A high-only accepted proposal is impossible in this scalar package order: if a package is high enough for the high type, it is also high enough for the low type under the stated domain. Within each accepted region, any package strictly above the relevant threshold is dominated for the weak proposer by lowering  $y$ . Any weak-voter offer strictly above the relevant continuation value is also dominated. Hence the only payoff-relevant pure threshold candidates with accepted agreements are pooling and low-only. Deliberate failure gives the continuation candidate. The informative-failure lemma below rules out a profitable fourth candidate in which the proposer uses a rejected ballot only to extract information before Round 2.

### 7.4.3 Candidate P: pooling at the high threshold

The pooling proposal is

$$y_P = a_1, \quad z_j = c(\mu) \quad \text{for all } j \neq i. \quad (\text{P-proposal})$$

It is physically feasible at belief  $\mu$  iff

$$a_1 + (m-1)c(\mu) \leq 1. \quad (\text{P-feas})$$

It is an admissible pooling voting assessment iff (P-feas) holds and  $a_1 \geq a_0(1)$ .

The proposed strategy profile is: both types of H vote yes; every non-proposing weak state votes yes.

Posterior beliefs after public vote vectors are:

$$\nu_P(v) = \begin{cases} \mu, & H \text{ votes yes,} \\ 1, & H \text{ votes no.} \end{cases} \quad (\text{P-beliefs})$$

The all-yes vector is on path and the proposal passes. Vectors with H voting yes and at least one weak no vote are off path but preserve the prior because H's prescribed yes vote is pooling. Vectors with H voting no are off path; the belief assigns probability one to the high type.

Weak voters' constraint is

$$z_j \geq c(\mu). \quad (\text{P-W})$$

If a weak voter votes yes, the proposal passes and it receives  $z_j$ . If it unilaterally votes no while all others follow the proposed strategy, the proposal fails, H's yes vote is uninformative, the posterior is  $\mu$ , and the voter receives  $c(\mu)$ . With  $z_j = c(\mu)$ , weak voters accept in indifference.

H's constraints are

$$a_1 + b_1 = \beta d_1 = \beta D_1(1), \quad (\text{P-H1})$$

so the high type accepts by the weak-acceptance convention, and

$$a_1 + b_0 \geq \beta D_0(1), \quad \text{equivalently} \quad a_1 \geq a_0(1), \quad (\text{P-H0})$$

so the low type also accepts. The latter is implied by the R1 separation domain.

The recognized weak proposer's pooling payoff is

$$\Pi_P^U(\mu) = 1 - a_1 - (m-1)c(\mu). \quad (\text{P-payoff})$$

**Voting-continuation proof.** Given the beliefs in (P-beliefs), each weak voter is indifferent between yes and no at  $z_j = c(\mu)$  and therefore votes yes by the weak-acceptance convention. The

high type of H is indifferent between acceptance and rejection at  $y = a_1$  and votes yes by the same convention. The low type weakly prefers acceptance because  $a_1 \geq a_0(1)$ . The proposer pays the minimum concession and minimum weak-voter offers consistent with these constraints, so no lower-cost pooling proposal exists. Beliefs on the all-yes path are Bayesian; off-path beliefs are explicitly specified. Therefore the pooling proposal is an admissible voting-continuation assessment whenever (P-feas) holds and  $a_1 \geq a_0(1)$ . It is selected in the baseline value only when the proposer-optimality condition stated below also selects pooling.

#### 7.4.4 Candidate L: low-only acceptance

The low-only proposal is

$$y_L = a_0(1), \quad z_j = c(0) \quad \text{for all } j \neq i. \quad (\text{L-proposal})$$

It is feasible iff

$$a_0(1) + (m-1)c(0) \leq 1, \quad (\text{L-feas})$$

and it requires the strict separation condition  $a_0(1) < a_1$ . The proposed strategy profile is: low H votes yes, high H votes no, and every non-proposing weak state votes yes.

Posterior beliefs after public vote vectors are:

$$\nu_L(v) = \begin{cases} 0, & H \text{ votes yes,} \\ 1, & H \text{ votes no.} \end{cases} \quad (\text{L-beliefs})$$

Thus the all-yes vector passes and reveals the low type. The vector in which H votes no and all weak states vote yes is on path in the high state, fails, and reveals the high type. Off-path vectors inherit the same rule: histories with H yes have posterior zero, and histories with H no have posterior one.

Weak voters' constraint is

$$z_j \geq c(0). \quad (\text{L-W})$$

To see this, compare a weak voter's yes and no payoffs given the proposed strategy profile. If the voter votes yes, it receives  $z_j$  in the low state and continuation  $c(1)$  in the high state, because high H rejects. If the voter votes no, the proposal fails in both states; the posterior is zero after low H's yes vote and one after high H's no vote. Hence the voter votes yes iff

$$(1 - \mu)z_j + \mu c(1) \geq (1 - \mu)c(0) + \mu c(1),$$

which reduces to  $z_j \geq c(0)$  when  $\mu < 1$ . At  $\mu = 1$ , weak voters are not payoff-pivotal for the on-path high-state rejection and accept weakly under the same offer convention.

H's constraints are

$$a_0(1) + b_0 = \beta D_0(1), \quad (\text{L-H0})$$

so the low type accepts by the weak-acceptance convention, and

$$a_0(1) + b_1 < \beta d_1, \quad \text{equivalently} \quad a_0(1) < a_1, \quad (\text{L-H1})$$

so the high type strictly rejects.

The recognized weak proposer's low-only payoff is

$$\Pi_L^U(\mu) = (1 - \mu)\{1 - a_0(1) - (m - 1)c(0)\} + \mu c(1). \quad (\text{L-payoff})$$

The second term is the proposer's continuation payoff after high H rejects and the public vote vector reveals posterior one.

**Voting-continuation proof.** Given (L-beliefs), each weak voter is exactly compensated for the continuation it can induce in the low state and is unaffected in the high state, so voting yes is sequentially rational. Low H is indifferent between acceptance and mimicking high rejection at posterior one and votes yes by the weak-acceptance convention. High H strictly prefers rejection because  $a_0(1) < a_1$ . The posterior after H's on-path no vote is one by Bayes, and the posterior after H's on-path yes vote is zero by Bayes. Off-path beliefs are explicitly stated in (L-beliefs). The proposer uses the minimum concession and weak-voter offers consistent with low-only acceptance. Therefore the low-only proposal is an admissible voting-continuation assessment whenever (L-feas) and  $a_0(1) < a_1$  hold. It is selected in the baseline value only when the proposer-optimality condition stated below also selects low-only.

#### 7.4.5 Candidate R: no-information rejection/continuation

The continuation proposal designates one non-proposing weak state  $r$  as a strict rejecter and sets

$$y_R = a_1, \quad z_j = 0 \quad \text{for all } j \neq i. \quad (\text{R-proposal})$$

It is feasible when  $a_1 \leq \bar{y} \leq 1$ . It is admissible as a strict rejection assessment at beliefs with  $c(\mu) > 0$ . The proposed voting strategy is: both types of H vote yes, the designated weak state  $r$  votes no, and all other non-proposing weak states vote yes. The proposal fails under unanimity because one weak state votes no.

The posterior after every failed vote vector following this proposal is

$$\nu_R(v) = \mu. \quad (\text{R-beliefs})$$

On path, H's yes vote is pooling and weak votes carry no private information, so this posterior is Bayesian. Off path, the same belief rule keeps unilateral vote deviations from changing continuation beliefs.

The designated weak state  $r$  strictly prefers no when  $c(\mu) > 0$ . If it votes no, the proposal fails and it receives  $c(\mu)$ . If it unilaterally votes yes while all others follow the proposed strategy, the proposal passes and it receives  $z_r = 0$ . All other non-proposing weak states are non-pivotal because  $r$  votes no; they vote yes by the non-pivotal tie-breaking rule.

H is not outcome-pivotal because  $r$  rejects. Under (R-beliefs), a unilateral H deviation from yes to no does not change the posterior, so both types of H are sequentially rational in voting yes. The choice  $y_R = a_1$  also means that if the weak rejection were removed, the high type would be at its acceptance threshold and the low type would weakly accept.

The recognized weak proposer's continuation payoff is

$$\Pi_R^U(\mu) = c(\mu). \quad (\text{R-payoff})$$

**Voting-continuation proof.** The on-path rejected vote vector preserves the prior by Bayes. The designated weak rejecter strictly prefers no when  $c(\mu) > 0$ . The other weak voters' yes votes are sequentially rational because their unilateral votes do not affect the outcome. H's yes vote is sequentially rational because H is not outcome-pivotal and the stated off-path beliefs make vote deviations payoff-irrelevant. The proposal therefore implements no-information continuation as an admissible voting-continuation assessment whenever  $a_1 \leq \bar{y}$  and  $c(\mu) > 0$ . It is selected in the baseline value only when the proposer-optimality condition stated below also selects rejection. This candidate is not a uniqueness claim about the voting subgame after a deliberately unacceptable proposal.

#### 7.4.6 Lemma: informative rejected ballots do not add a profitable candidate

An informative rejected ballot would try to make the proposal fail in both states while inducing different H votes, so that the public vote vector changes the posterior before Round 2. Such a ballot cannot strictly improve on the no-information continuation candidate under the maintained domain.

To see this, suppose a rejected ballot separates H's types. Since weak voters do not observe  $\theta$ , only H's vote can carry type information. In any separating rejected ballot, one H action must induce posterior zero and the other must induce posterior one. The low type's continuation payoff from revealing the low posterior is

$$\beta D_0(0) = \beta d_0.$$

The low type's payoff from mimicking the high-posterior vote is

$$\beta D_0(1) \geq \beta D_0(0),$$

with strict inequality whenever  $\tau_1 < 1$ . Therefore the low type does not strictly prefer to reveal itself in a rejected ballot; when the inequality is strict, separation is not incentive-compatible. When  $\tau_1 = 1$ , the inequality is an equality and the Round-2 weak-proposer value is linear:

$$p_2(\nu) = (1 - \nu)(1 - \tau_0) \quad \text{for all } \nu \in [0, 1].$$

Thus even if an informative rejected ballot is sustained by indifference at the boundary, it gives the weak proposer

$$(1 - \mu)c(0) + \mu c(1) = c(\mu),$$

exactly the no-information continuation payoff. Pooling rejected ballots also preserve the prior and give  $c(\mu)$ . Hence rejected ballots add no candidate payoff above  $\Pi_R^U(\mu)$ .

Status: proved under the fixed-pie R2 continuation values and simultaneous public vote records.

#### 7.4.7 Weak-vote-passive pure-threshold selected-value rule

The preceding subsections establish the voting-continuation assessment for each candidate under the weak-vote-passive assessment. The selected baseline value also requires proposer optimality within the pure-threshold candidate set.

For each  $\mu$ , define the admissible candidate set

$$\mathcal{K}(\mu) \subseteq \{P, L, R\}$$

as follows:  $P \in \mathcal{K}(\mu)$  iff (P-feas) and  $a_1 \geq a_0(1)$  hold;  $L \in \mathcal{K}(\mu)$  iff (L-feas) and  $a_0(1) < a_1$  hold; and  $R \in \mathcal{K}(\mu)$  iff  $a_1 \leq \bar{y}$  and  $c(\mu) > 0$ . Inadmissible candidates receive payoff  $-\infty$ . Let

$$\bar{\Pi}^U(\mu) = \max_{k \in \mathcal{K}(\mu)} \Pi_k^U(\mu). \quad (\text{R1-max})$$

The proposer-optimal candidates are

$$\mathcal{A}(\mu) = \{k \in \mathcal{K}(\mu) : \Pi_k^U(\mu) = \bar{\Pi}^U(\mu)\}. \quad (\text{R1-argmax})$$

If  $\mathcal{A}(\mu)$  contains more than one candidate, the maintained tie-breaking rule selects the candidate that minimizes H's expected payoff under the prior  $\mu$ . These expected H payoffs are

$$H_P(\mu) = (1 - \mu)(a_1 + b_0) + \mu(a_1 + b_1),$$

$$H_L(\mu) = (1 - \mu)(a_0(1) + b_0) + \mu\beta d_1,$$

and

$$H_R(\mu) = (1 - \mu)\beta D_0(\mu) + \mu\beta d_1.$$

Thus the baseline selected candidate is

$$k^*(\mu) \in \arg \min_{k \in \mathcal{A}(\mu)} H_k(\mu). \quad (\text{R1-tie-break})$$

For any selected  $k^*(\mu)$ , the strategy profile is the corresponding candidate proposal and voting strategy above, with the stated posterior map after the public vote vector. The candidate proof gives responder sequential rationality and Bayesian consistency on path. Off-path histories are disciplined by the weak-vote-passive assessment above. The argmax condition gives proposer optimality. The informative-failure lemma rules out rejected-ballot deviations that would use public voting only to

generate a strictly better posterior payoff. Therefore the selected assessment delivers a selected R1 value within the pure threshold class generated by the fixed-pie scalar package order. This is the baseline value under the maintained weak-vote-passive assessment, not a global exhaustion result over unrestricted PBEs.

#### 7.4.8 Weak-vote-passive pure-threshold selected value under R1 unanimity

For each prior  $\mu$ , define invalid candidate values as  $-\infty$ . The recognized weak proposer's selected weak-vote-passive pure-threshold R1 unanimity value is

$$P_{1,prop}^U(\mu) = \bar{\Pi}^U(\mu), \quad (\text{R1-U-value})$$

where admissibility is defined in  $\mathcal{K}(\mu)$  above. The corresponding proposer-recognition component for a symmetrically anonymous weak state is

$$W_{1,prop}^U(\mu) = \frac{P_{1,prop}^U(\mu)}{m},$$

because each weak state is recognized with probability  $1/m$  under  $\pi_H = 0$ . This is not yet the collective entry payoff of a representative weak state whenever non-proposing weak voters receive positive continuation offers. The entry section below adds those voter payments back and uses total weak payoff divided by  $m$ .

The important correction relative to any informal low-only intuition is that the low type's R1 acceptance threshold is not  $\beta d_0 - b_0$ . In a separating low-only proposal, a low type that rejects mimics the high type in the public vote record and receives the high-posterior R2 continuation value. The correct low-only concession is therefore  $a_0(1) = \beta D_0(1) - b_0$ . If this reaches  $a_1$ , low-only separation is blocked.

Status ledger for R1 unanimity:

| Object                                | Status                                  | Notes                                                                                                                                                                                                                  |
|---------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1 primitives                         | Specified                               | Players, types, proposer identity, proposal space, voting rule, payoffs, and continuation values are explicit.                                                                                                         |
| Weak-vote-passive candidate reduction | Established within pure threshold class | Accepted proposals reduce to pooling or low-only; informative rejected ballots do not improve on no-information continuation under the maintained assessment. This is not a characterization of all unrestricted PBEs. |



| Object                                       | Status                                                 | Notes                                                                                                                                                                                      |
|----------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pooling candidate                            | Derived and checked under weak-vote-passive assessment | Admissible voting-continuation assessment under (P-feas) and $a_1 \geq a_0(1)$ ; selected when it solves the argmax/tie-break problem.                                                     |
| Low-only candidate                           | Derived and checked under weak-vote-passive assessment | Admissible voting-continuation assessment under (L-feas) and strict $a_0(1) < a_1$ ; blocked when $a_0(1) \geq a_1$ ; selected when it solves the argmax/tie-break problem.                |
| Rejection/continuation candidate             | Derived and checked under weak-vote-passive assessment | Admissible no-information continuation assessment when $a_1 \leq \bar{y}$ and $c(\mu) > 0$ ; selected when it solves the argmax/tie-break problem.                                         |
| Weak-vote-passive pure-threshold selection   | Derived and checked                                    | Selected candidates maximize the proposer payoff among admissible candidates; ties minimize H's expected payoff.                                                                           |
| Global characterization of all standard PBEs | Not claimed                                            | Standard PBE permits arbitrary off-path beliefs after weak-voter deviations. The paper uses the weak-vote-passive assessment as a maintained baseline interpretation, not as a refinement. |
| Numerical and algebraic verification         | Checked                                                | See <code>scripts/verify_relative_package_R1_piH</code>                                                                                                                                    |

## 7.5 Majority

Let

$$q = \lfloor N/2 \rfloor + 1, \quad k = q - 1.$$

The recognized weak proposer is counted as voting yes, so under majority it needs  $k$  additional yes votes.

### 7.5.1 R2 majority

In terminal Round 2, weak voters' outside payoff is zero. The proposer can set  $y = 0$ , offer zero to  $k$  weak voters, and pass the proposal without H. Any  $y > 0$  is costly to the weak coalition and is not needed for passage. Thus the recognized weak proposer's terminal majority value is

$$P_{2,prop}^M = 1.$$

The representative weak state's terminal majority value is

$$W_2^M = \frac{1}{m}.$$

If excluded under majority, H receives its external payoff  $d_\theta$ , outside the weak coalition's institutional surplus:

$$C_{H,2}^M(\theta) = d_\theta.$$

### 7.5.2 R1 majority

The public continuation value of a weak voter after R1 rejection is

$$c_M = \beta W_2^M = \frac{\beta}{m}.$$

A no-H majority proposal excludes H from the agreement, sets  $y = 0$ , offers  $c_M$  to  $k$  weak voters, and offers zero to all other states. H may still cast a public vote, but its vote is not pivotal and does not bring H inside the agreement or give it the agreement benefit  $b_\theta$ . The proposal is accepted by the selected weak voters by the Baron–Ferejohn weak-acceptance convention. The recognized weak proposer's payoff is

$$P_{1,noH}^M = 1 - kc_M = 1 - \frac{(q-1)\beta}{m}.$$

Deliberate majority rejection gives the proposer  $c_M$ . The accepted no-H proposal weakly dominates rejection because

$$P_{1,noH}^M - c_M = 1 - (k+1)c_M = 1 - \frac{q\beta}{m} \geq 0,$$

using  $q \leq m$  for  $N \geq 3$  and  $\beta \leq 1$ .

Now check whether the proposer can do better by buying H's vote and one fewer weak vote. Any majority package that includes H and still buys  $k$  weak votes is weakly dominated by dropping the concession to H, since H's vote is not needed for passage. Any H-including package that sets  $y$  strictly inside an acceptance region is dominated by lowering  $y$  to the relevant threshold. Thus the only potentially profitable H-including alternatives are pooling at the high threshold and low-only at the low threshold, each with  $k-1$  weak voters.

If H is made pivotal in an R1 majority coalition, its type-specific threshold is

$$a_\theta^M = \beta C_{H,2}^M(\theta) - b_H(\theta) = \beta d_\theta - b_H(\theta).$$

Assume the R1 majority threshold domain

$$0 \leq a_0^M < a_1^M \leq \bar{y}.$$

A pooling H-including majority proposal costs  $a_1^M + (k-1)c_M$ . It is not better than the no-H proposal iff

$$a_1^M \geq c_M.$$

A low-only H-including majority proposal costs  $a_0^M + (k-1)c_M$  when  $\theta = 0$ , but fails and yields continuation value  $c_M$  to the proposer when  $\theta = 1$ . Its expected payoff is

$$P_{1,LH}^M(\mu) = (1-\mu)\{1 - a_0^M - (k-1)c_M\} + \mu c_M.$$

Relative to the no-H proposal,

$$P_{1,LH}^M(\mu) - P_{1,noH}^M = c_M - a_0^M + \mu(a_0^M + kc_M - 1). \quad (\text{majority-low-H-gap})$$

Since  $q = k+1 \leq m$  for  $N \geq 3$ , and  $\beta \leq 1$ ,

$$(k+1)c_M = \frac{q\beta}{m} \leq 1. \quad (\text{quota-bound})$$

Therefore, if

$$a_0^M \geq c_M. \quad (\text{no-cheap-H})$$

then  $P_{1,LH}^M(0) - P_{1,noH}^M \leq 0$  and  $P_{1,LH}^M(1) - P_{1,noH}^M = (k+1)c_M - 1 \leq 0$ . The gap is linear in  $\mu$ , so the low-only H-including proposal is weakly dominated by the no-H proposal for every  $\mu \in [0, 1]$ .

Because  $a_1^M > a_0^M$ , condition (no-cheap-H) also implies  $a_1^M \geq c_M$ , so pooling purchases of H's vote are not profitable. Under this condition, the R1 majority no-H path has  $y = 0$ , a weak-state winning coalition, and no screening of H.

The condition is also necessary for uniform no-screening. If  $a_0^M < c_M$ , then at  $\mu = 0$

$$P_{1,LH}^M(0) - P_{1,noH}^M = c_M - a_0^M > 0.$$

By continuity, a low-only H-including proposal strictly beats the no-H proposal for sufficiently low  $\mu$ , when the high type is unlikely. In that region, majority itself has a screening motive: the weak proposer buys low H as a cheap substitute for one weak voter and lets the proposal fail when H is high. The cutoff for this majority-screening region is

$$\mu_M^H = \frac{c_M - a_0^M}{1 - a_0^M - kc_M}$$

whenever  $a_0^M < c_M$ . The quota bound above implies  $\mu_M^H \in (0, 1]$ .

Thus condition (no-cheap-H) is not a verbal regularity condition. It is the payoff condition under which the no-H majority proposal weakly dominates the H-including majority proposals considered here for all beliefs. The voting-continuation assessment below supplies the associated strategy and belief map for these payoff-relevant threshold proposals.

The recognized weak proposer's R1 majority value is therefore

$$P_{1,prop}^M = 1 - \frac{(q-1)\beta}{m},$$

and the symmetric ex ante representative weak-state value is

$$W_1^M = \frac{1}{m}.$$

Status: payoff path derived under the simultaneous public voting protocol and the no-cheap-H condition. The derivation is checked in `scripts/verify_relative_package_majority_piH0.R`.

### 7.5.3 Majority voting-continuation assessment

The majority payoff comparison above can be supported by the following voting-continuation assessment for the payoff-relevant threshold proposals. This is not a characterization of all majority voting PBEs after arbitrary proposals; it is the strategy and belief map behind the no-screening benchmark and the cheap-H diagnostic branch.

For the no-H majority proposal, the proposer sets  $y = 0$ , offers  $c_M$  to exactly  $k = q - 1$  weak voters, and offers zero to all other states. The proposer votes yes. The  $k$  selected weak voters vote yes. Non-selected weak voters vote no. Both types of H vote no. The proposal passes with exactly  $q$  yes votes from the proposer and the selected weak voters; H is outside the agreement and receives  $d_\theta$ . Since both types of H cast the same non-pivotal vote, H's vote carries no type information on this path. If a selected weak voter unilaterally deviates to no, the proposal fails, the public history contains no type-dependent H vote, beliefs remain at the prior, and the weak voter receives  $c_M$ . Thus selected weak voters accept at  $z_j = c_M$  by the weak-acceptance convention. Non-selected weak voters are not outcome-pivotal on this path and receive zero whether they vote yes or no.

For an H-including pooling majority proposal, the proposer sets  $y = a_1^M$ , offers  $c_M$  to  $k - 1$  weak voters, and counts on H as the remaining coalition member. Both H types vote yes. The  $k - 1$  selected weak voters vote yes; non-selected weak voters vote no. H's yes vote pools the two types and therefore does not update beliefs. Selected weak voters receive their continuation value if the proposal passes and receive the same value after a unilateral no deviation that makes the proposal fail, so their yes votes are sequentially rational. High H accepts at its threshold  $a_1^M + b_1 = \beta d_1$ , and low H weakly accepts because  $a_1^M > a_0^M$ .

For an H-including low-only majority proposal, the proposer sets  $y = a_0^M$ , offers  $c_M$  to  $k - 1$  weak voters, and again counts on H as the remaining coalition member. Low H votes yes and high H votes no. The proposal passes only in the low state. The vote vector is type-revealing on path: histories with H voting yes have posterior zero, and histories with H voting no have posterior one. Selected weak voters vote yes because in the low state their deviation would induce failure and continuation value  $c_M$ , while in the high state the proposal fails regardless. Low H accepts at  $a_0^M + b_0 = \beta d_0$ , and high H strictly rejects because  $a_0^M < a_1^M$ .

Under no-cheap-H, the no-H proposal weakly dominates both H-including threshold proposals for every belief, so the no-screening majority path is the selected benchmark path. If no-cheap-H fails, the same voting-continuation assessment identifies the low-belief region where the low-only H-including proposal is payoff-superior; that region is a majority-screening extension, not part of the clean no-screening contrast used in the baseline institutional comparison.

#### 7.5.4 Majority without H when weak-only surplus is $\rho < 1$

The baseline majority benchmark assumes that weak states can exclude H and still preserve the full weak-coalition surplus normalized to one. This is a deliberately favorable benchmark for majority, but it is substantively strong in the OPEC interpretation. If Saudi Arabia is excluded, the remaining producers may not be able to preserve the same cartel surplus.

As a stress test, let the surplus from a majority agreement that excludes H be

$$\rho \in (0, 1],$$

while an agreement that includes H preserves the full normalized surplus one. This is not a new empirical calibration. It is a reduced-form way to check how much weak-only surplus the majority benchmark needs in order to remain a clean no-screening contrast to unanimity.

In terminal majority, a no-H proposal gives the recognized weak proposer  $\rho$ . An H-including terminal low-only proposal gives at most  $1 - \tau_0$ . Therefore the no-H terminal continuation is selected uniformly for all beliefs if

$$\rho \geq 1 - \tau_0. \quad (\text{rho-terminal})$$

Under this condition, the weak voter's public R1 continuation value on the no-H majority path is

$$c_M(\rho) = \frac{\beta\rho}{m}.$$

The R1 no-H majority proposal gives the recognized weak proposer

$$P_{1,noH}^M(\rho) = \rho - kc_M(\rho).$$

An H-including low-only proposal gives

$$P_{1,LH}^M(\mu; \rho) = (1 - \mu)\{1 - a_0^M - (k - 1)c_M(\rho)\} + \mu c_M(\rho).$$

Relative to the no-H proposal,

$$P_{1,LH}^M(\mu; \rho) - P_{1,noH}^M(\rho) = 1 - \rho + c_M(\rho) - a_0^M + \mu\{a_0^M + kc_M(\rho) - 1\}. \quad (\text{rho-majority-low-H-gap})$$

The low-only H-including proposal is therefore not profitable for any belief if

$$a_0^M \geq 1 - \rho + \frac{\beta\rho}{m}. \quad (\text{rho-no-cheap-H})$$

Equivalently,

$$\rho \geq \frac{1 - a_0^M}{1 - \beta/m}. \quad (\text{rho-r1-min})$$

Combining terminal and R1 requirements, the clean no-H majority benchmark is preserved if

$$\rho \geq \bar{\rho}^M \equiv \max \left\{ 1 - \tau_0, \frac{1 - a_0^M}{1 - \beta/m} \right\}. \quad (\text{rho-clean-majority})$$

If  $\rho < \bar{\rho}^M$ , the majority benchmark no longer cleanly excludes H. Either the terminal majority continuation itself may want to include H for some beliefs, or, when the terminal no-H condition still holds but the R1 condition fails, a weak proposer can gain by buying low-type H as a cheap coalition partner at sufficiently low beliefs. In the latter case the R1 screening cutoff is

$$\mu_M^H(\rho) = \frac{1 - \rho + \beta\rho/m - a_0^M}{1 - a_0^M - k\beta\rho/m}, \quad (\text{rho-majority-screening-cutoff})$$

whenever the numerator is positive.

For the working numerical illustration,

$$N = 13, \quad \beta = 0.9, \quad d_0 = 0.19, \quad d_1 = 0.285, \quad b_0 = b_1 = 0,$$

the stress-test thresholds are

$$1 - \tau_0 = 0.81, \quad \frac{1 - a_0^M}{1 - \beta/m} = 0.896216.$$

Thus the no-H majority benchmark survives this reduced-surplus stress test only if weak states retain at least about 89.6% of the full surplus when they exclude H. The weaker entry-nesting condition in the same numerical example is only

$$\rho \geq mV_W^U = 0.7435,$$

so the binding issue is not entry nesting. It is whether majority can still be treated as a no-screening institution when excluding H substantially reduces the value of coordination.

Status: reduced-surplus majority stress test derived as an interpretive/OPEC extension, not as an empirical calibration. The thresholds and example values are checked in `scripts/verify_relative_package_rho`.

### 7.5.5 OPEC interpretation discipline

The current OPEC use should be framed as an interpretation of the mechanism, not as evidence for the numerical example. The formal baseline intentionally sets  $\pi_H = 0$ : Saudi Arabia is pivotal and privately informed, but it is not the Round-1 agenda setter. This stacks the benchmark against agenda-power explanations and isolates informational power through pivotality.

Under this interpretation, the parameters in the working numerical example are not estimates. The safe mapping is:

| Model object                | OPEC interpretation                                                                                                                                                     | Current evidentiary status                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| $d_\theta$ or $o_H(\theta)$ | Saudi Arabia's outside/continuation value from rejecting a proposed package, reflecting spare capacity, fiscal needs, opportunity cost, and expected market conditions. | Primitive in this proof pass; not empirically estimated.       |
| $b_H(\theta)$               | Direct value to Saudi Arabia from participating in an agreement apart from the concession $y$ .                                                                         | Primitive in this proof pass; not empirically estimated.       |
| $y$                         | Relative institutional concession: quota share, exemption, monitoring advantage, production flexibility, or enforcement term.                                           | Conceptual mapping only.                                       |
| $\rho$                      | Fraction of coordination surplus retained by a majority coalition that excludes Saudi Arabia.                                                                           | Reduced-form stress-test parameter; not an empirical estimate. |
| $\mu$                       | Weak members' belief that Saudi Arabia has a high participation threshold.                                                                                              | Belief parameter; not measured in the numerical illustration.  |

Therefore the paper-ready OPEC language should make only three claims unless a separate empirical calibration is added. First, Saudi Arabia plausibly fits the role of a pivotal privately informed actor because its participation threshold depends on quantities other members do not fully observe. Second, excluding Saudi Arabia under majority may reduce the surplus available to the remaining coalition, which the model represents by  $\rho < 1$ . Third, the numerical threshold  $\bar{\rho}^M$  shows how demanding the clean no-H majority benchmark is in the illustrative parameterization; it does not estimate the historical value of majority rule in OPEC.

If the manuscript later uses calibration language, the missing empirical inputs must be supplied outside this proof pass: a source for Saudi outside-option or opportunity-cost variation, a source for quota/concession magnitudes, a source for the value of coordination with and without Saudi

participation, and a transparent way to map these quantities into  $d_\theta$ ,  $b_H(\theta)$ ,  $\rho$ , and beliefs  $\mu$ . Until then, the correct label is interpretive application or illustrative stress test.

### 7.5.6 Initial-recognition agenda-power bounds: $\pi_H > 0$

The main baseline sets  $\pi_H = 0$  so that H is pivotal under unanimity but does not have formal proposal power. This isolates informational power through pivotality from agenda power.

This extension changes only the initial Round-1 recognition probability while holding the continuation protocol at the  $\pi_H = 0$  baseline derived above. It is therefore an initial-recognition agenda-power stress test, not a recursive solution of the full two-round game with  $\pi_H > 0$  in every continuation. If  $\pi_H > 0$  also applies after rejection, then the weak-proposer continuation values themselves must be rederived and the bounds below are not a complete solution of that recursive game.

Under this initial-recognition interpretation, the expected payoff of any actor class under rule  $R \in \{U, M\}$  can be decomposed as

$$V_i^R(\mu; \pi_H) = (1 - \pi_H)V_i^{R,W}(\mu) + \pi_H V_i^{R,H}(\mu), \quad (\text{piH-decomposition})$$

where  $V_i^{R,W}$  is the weak-proposer value derived above and  $V_i^{R,H}$  is the value in the H-proposer branch.

This document does not try to solve the H-proposer signaling branch as a unique payoff function. That branch can involve pooling, separating, or off-path belief-dependent behavior. Instead, the extension records selection-free bounds. Under fixed-pie relative packages, any accepted package has  $y \leq 1$ , and H can always induce rejection and receive at least its discounted terminal disagreement payoff. Thus, for type  $\theta$ ,

$$\beta d_\theta \leq V_H^{R,H}(\theta, \mu) \leq 1 + b_\theta. \quad (\text{piH-H-bounds})$$

For an anonymous representative weak state, interpreted as the average payoff across the  $m$  weak states before identities are assigned, the fixed-pie normalization implies

$$0 \leq V_W^{R,H}(\mu) \leq \frac{1}{m}. \quad (\text{piH-W-bounds})$$

Combining these bounds with (piH-decomposition) gives immediate interval payoffs for any  $\pi_H \in [0, 1]$ . The intervals collapse to the exact baseline payoffs when  $\pi_H = 0$ . They widen as formal agenda power for H increases, which is the substantive point: once H can be the proposer, agenda power and pivotality are no longer cleanly separated without solving the signaling branch or imposing an additional selection rule.

For the working numerical illustration, the bounds are reported in `tables/relative_package_piH_bounds.csv` and checked in `scripts/verify_relative_package_piH_bounds.R`. This is a robustness and scope exercise, not a replacement for the clean  $\pi_H = 0$  theorem architecture and not an exact solution of the full recursive  $\pi_H > 0$  game.



## 7.6 Entry and Institutional Choice

Status: proved for weak-state entry/nesting under the selected pure-threshold R1 unanimity value and the no-cheap-H majority condition. Checked in `scripts/verify_relative_package_entry_nesting_piH0.R`.

### 7.6.1 Weak-state entry payoff

Entry by weak states is collective and all-or-nothing. Therefore the entry payoff must include all weak-state payoffs generated by the selected R1 equilibrium path, not only the residual kept by the recognized proposer.

Let  $k^*(\mu) \in \{P, L, R\}$  denote the selected weak-vote-passive pure-threshold R1 unanimity candidate from (R1-tie-break). The total expected weak payoff under each candidate is:

$$S_P^U(\mu) = 1 - a_1, \quad (\text{entry-P})$$

because pooling passes in both states and the concession  $a_1$  is the only loss from the fixed weak pie;

$$S_L^U(\mu) = (1 - \mu)(1 - a_0(1)) + \mu\beta p_2(1), \quad (\text{entry-L})$$

because the low-only proposal passes in the low state and continues to R2 with posterior one in the high state; and

$$S_R^U(\mu) = \beta p_2(\mu), \quad (\text{entry-R})$$

because deliberate rejection preserves the prior and gives the weak coalition its discounted R2 value.

The representative weak-state entry payoff under unanimity is

$$V_W^U(\mu) = \frac{S_{k^*(\mu)}^U(\mu)}{m}. \quad (\text{VW-U-entry})$$

Under majority, impose the no-cheap-H condition derived above,

$$a_0^M = \beta d_0 - b_0 \geq c_M = \frac{\beta}{m}. \quad (\text{entry-no-cheap-H})$$

Then the no-H majority proposal is selected for every belief, total weak payoff is one, and

$$V_W^M(\mu) = \frac{1}{m} \quad \text{for all } \mu \in [0, 1]. \quad (\text{VW-M-entry})$$

### 7.6.2 Entry sets and nesting

Let  $\chi \geq 0$  be the per-weak-state entry or formation cost. Define

$$F_R(\chi) = \{\mu \in [0, 1] : V_W^R(\mu) \geq \chi\}, \quad R \in \{U, M\}. \quad (\text{entry-set})$$

The entry condition is therefore explicit:

$$\mu \in F_U(\chi) \iff S_{k^*(\mu)}^U(\mu) \geq m\chi, \quad (\text{FU-condition})$$

while

$$F_M(\chi) = \begin{cases} [0, 1], & \chi \leq 1/m, \\ \emptyset, & \chi > 1/m. \end{cases} \quad (\text{FM-condition})$$

**Proposition: weak-state nesting.** Under the fixed-pie relative-package threshold domain, the weak-vote-passive pure-threshold R1 unanimity characterization, and the no-cheap-H majority condition, majority weakly dominates unanimity for weak-state entry:

$$V_W^U(\mu) \leq V_W^M(\mu) = \frac{1}{m} \quad \text{for every } \mu \in [0, 1]. \quad (\text{weak-payoff-dominance})$$

Consequently,

$$F_U(\chi) \subseteq F_M(\chi) \quad \text{for every entry cost } \chi \geq 0. \quad (\text{nesting})$$

**Proof.** For pooling,  $S_P^U(\mu) = 1 - a_1 \leq 1$  because  $a_1 \geq 0$ . For low-only,

$$S_L^U(\mu) = (1 - \mu)(1 - a_0(1)) + \mu\beta p_2(1) \leq (1 - \mu) + \mu = 1,$$

because  $a_0(1) \geq 0$ ,  $p_2(1) \leq 1$ , and  $\beta \leq 1$ . For rejection,

$$S_R^U(\mu) = \beta p_2(\mu) \leq 1.$$

The selected candidate  $k^*(\mu)$  is one of these three candidates, so  $S_{k^*(\mu)}^U(\mu) \leq 1$  and  $V_W^U(\mu) \leq 1/m = V_W^M(\mu)$ . The set inclusion follows immediately from the definition of  $F_R(\chi)$ .

The strict version is also transparent. If the selected unanimity path destroys strictly positive weak surplus—for example  $a_1 > 0$  under pooling,  $a_0(1) > 0$  on the accepted low-only path, or  $\beta p_2(\mu) < 1$  under rejection—then the majority payoff is strictly larger at that belief.

### 7.6.3 Conditional institutional comparison for H

Weak-state nesting does not by itself imply that H always prefers unanimity. Under no-H majority, H receives its external payoff:

$$V_H^M(\mu) = (1 - \mu)d_0 + \mu d_1. \quad (\text{VH-M-entry})$$

Under unanimity, the selected R1 candidate gives

$$H_P(\mu) = (1 - \mu)(a_1 + b_0) + \mu(a_1 + b_1),$$

$$H_L(\mu) = (1 - \mu)(a_0(1) + b_0) + \mu\beta d_1,$$

and

$$H_R(\mu) = (1 - \mu)\beta D_0(\mu) + \mu\beta d_1.$$

Thus

$$V_H^U(\mu) = H_{k^*(\mu)}(\mu), \quad \Delta_H(\mu) = V_H^U(\mu) - V_H^M(\mu). \quad (\text{H-gap})$$

**Proposition: conditional institutional comparison for H.** Under the fixed-pie relative-package threshold domain, the weak-vote-passive pure-threshold R1 unanimity characterization, and the no-cheap-H majority condition, the hegemon's institutional ranking on the set where both rules form is completely determined by  $\Delta_H(\mu)$ :

$$\begin{aligned} \Delta_H(\mu) > 0 &\iff \text{H strictly prefers unanimity,} \\ \Delta_H(\mu) = 0 &\iff \text{H is indifferent,} \\ \Delta_H(\mu) < 0 &\iff \text{H strictly prefers majority.} \end{aligned} \quad (\text{H-ranking})$$

This is a conditional comparison, not a global dominance theorem. The sign of  $\Delta_H(\mu)$  depends on the selected unanimity candidate and on the primitive outside payoffs.

### 7.6.4 Complete institutional classification

The entry convention is weak: weak states form an institution under rule  $R$  when  $V_W^R(\mu) \geq \chi$ . Equality is therefore a formation tie, not a failure to form. The hegemon's institutional tie is kept separate: if  $\Delta_H(\mu) = 0$ , the classification records indifference rather than imposing an institutional tie-breaker.

For any fixed entry cost  $\chi \geq 0$ , define the following five sets:

$$\mathcal{C}_0(\chi) = [0, 1] \setminus F_M(\chi), \quad (\text{class-none})$$

$$\mathcal{C}_M(\chi) = F_M(\chi) \setminus F_U(\chi), \quad (\text{class-majority-only})$$

$$\mathcal{C}_U^H(\chi) = F_U(\chi) \cap \{\mu : \Delta_H(\mu) > 0\}, \quad (\text{class-both-H-U})$$

$$\mathcal{C}_T^H(\chi) = F_U(\chi) \cap \{\mu : \Delta_H(\mu) = 0\}, \quad (\text{class-both-H-tie})$$

and

$$\mathcal{C}_M^H(\chi) = F_U(\chi) \cap \{\mu : \Delta_H(\mu) < 0\}. \quad (\text{class-both-H-M})$$

**Corollary: complete weak-vote-passive institutional classification under  $\pi_H = 0$ .** Under the weak-vote-passive pure-threshold selected value and tie-breaking convention used in the R1 unanimity theorem, and under the conditions of the weak-state nesting proposition, the five sets above form a mutually exclusive and exhaustive partition of  $[0, 1]$ . Their substantive meaning is:

1.  $\mathcal{C}_0(\chi)$ : no rule forms;
2.  $\mathcal{C}_M(\chi)$ : only majority forms;
3.  $\mathcal{C}_U^H(\chi)$ : both rules form, and H strictly prefers unanimity;
4.  $\mathcal{C}_T^H(\chi)$ : both rules form, and H is indifferent;
5.  $\mathcal{C}_M^H(\chi)$ : both rules form, and H strictly prefers majority.

There is no sixth “only unanimity forms” case in the no-screening majority domain because  $F_U(\chi) \subseteq F_M(\chi)$  for every  $\chi$ . Since

$$F_M(\chi) = \begin{cases} [0, 1], & \chi \leq 1/m, \\ \emptyset, & \chi > 1/m, \end{cases}$$

the no-formation case is simply all beliefs when  $\chi > 1/m$ , and empty when  $\chi \leq 1/m$ . At  $\chi = 1/m$ , majority forms by the weak entry convention. At beliefs where  $V_W^U(\mu) = \chi$ , unanimity also forms by the same convention. These entry ties are distinct from hegemon indifference at  $\Delta_H(\mu) = 0$ .

**Proof.** The definitions split  $[0, 1]$  first by whether majority forms, then by whether unanimity also forms, and finally by the sign of  $\Delta_H(\mu)$  on the both-form set. The sets are therefore disjoint and cover  $[0, 1]$ . Weak-state nesting rules out  $F_U(\chi) \setminus F_M(\chi)$ , so no only-unanimity category can occur.

The classification is checked in `scripts/verify_relative_package_classification_piH0.R`, including formation ties, hegemon indifference ties, the no-cheap-H boundary, and  $\beta = 1$ .

### 7.6.5 Illustrative numerical classification

For the working numerical illustration

$$N = 13, \quad \beta = 0.9, \quad d_0 = 0.19, \quad d_1 = 0.285, \quad b_0 = b_1 = 0,$$

the majority no-cheap condition holds:

$$a_0^M = 0.171 > c_M = 0.075.$$

The selected R1 unanimity candidate is pooling for every  $\mu$ . Hence

$$V_W^U(\mu) = \frac{1 - 0.2565}{12} = 0.0619583, \quad V_W^M(\mu) = \frac{1}{12} = 0.0833333.$$

The minimum weak-state nesting gap is

$$\min_{\mu} \{V_W^M(\mu) - V_W^U(\mu)\} = 0.021375.$$

Therefore  $F_U(\chi) \subseteq F_M(\chi)$  for every  $\chi$ . The conditional H comparison is

$$\Delta_H(\mu) = 0.0665 - 0.095\mu,$$

so H weakly prefers unanimity for  $\mu \leq 0.7$  and majority for higher beliefs.

The illustrative institutional classification is therefore:

$$\begin{array}{ll} 0 \leq \chi \leq 0.0619583 : & \begin{cases} \text{both form and H strictly prefers unanimity,} & \mu < 0.7, \\ \text{both form and H is indifferent,} & \mu = 0.7, \\ \text{both form and H strictly prefers majority,} & \mu > 0.7; \end{cases} \\ 0.0619583 < \chi \leq 0.0833333 : & \text{only majority forms for every } \mu; \\ \chi > 0.0833333 : & \text{no rule forms for every } \mu. \end{array} \quad (\text{illustrative-classification})$$

At the lower boundary  $\chi = 0.0619583$ , unanimity forms because weak states are indifferent between forming and not forming. At  $\chi = 1/12$ , majority forms by equality, but unanimity does not form in this numerical illustration because  $V_W^U < 1/12$ .

A strict low-only-available diagnostic with  $b_0 = 0.03$  also satisfies no-cheap-H. In that case the selected unanimity path is rejection for  $\mu \in [0, 0.0164]$  on the numerical grid and pooling thereafter. The maximum unanimity weak entry payoff is 0.063, the majority payoff remains 0.0833333, and the minimum nesting gap is 0.0203333. In that diagnostic H prefers majority at very low beliefs, unanimity at intermediate beliefs, and majority again at high beliefs. This is a diagnostic robustness case, not an empirical calibration.

The local robustness check in `scripts/verify_relative_package_robustness_piH0.R` uses the interior low-only/pooling center

$$(\beta, t_0, t_1, o_0, o_1) = (0.9, 0.19, 0.285, 0.19, 0.265)$$

At this center, the script verifies strict margins for the threshold-order condition, no-cheap-H, entry nesting, positive and negative values of  $\Delta_H$ , low-only selection at  $\mu = 0$ , and pooling selection at  $\mu = 0.01$ . Because all candidate payoffs and thresholds are continuous in the primitive parameters wherever the threshold domain holds, these strict inequalities imply that there exists a nonempty open neighborhood around the center in which the same qualitative features persist. The reported margin table is `tables/relative_package_local_continuity_margins_piH0.csv`.

As a numerical check around that continuity argument, the script also verifies all  $3^5 = 243$  grid points in the box obtained by perturbing each of  $\beta, t_0, t_1, o_0, o_1$  by  $-0.0005, 0, 0.0005$ . Every point in this grid satisfies the core conditions, has low-only selected for some beliefs, has pooling selected for some beliefs, and preserves a strictly positive weak-entry nesting gap. The generated grid table is `tables/relative_package_open_neighborhood_piH0.csv`. These are still illustrative robustness checks, not empirical calibration.

### 7.6.6 Paper-ready theorem architecture

The following architecture is a candidate for paper transport after external audit. Main-text statements should give intuition and compact results; proofs should remain in the appendix, and the R1 result should be stated as a selected value under the maintained weak-vote-passive assessment.

| Result                                        | Type                | Claim                                                                                                                                                                                                                                                        | Status                                                                                                           |
|-----------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Terminal unanimity                            | Proposition/Theorem | Under $0 \leq \tau_0 < \tau_1 \leq \bar{y} \leq 1$ , Round 2 selects low-only for $\mu \leq \mu_2^*$ and pooling for $\mu > \mu_2^*$ , with $\mu_2^* = (\tau_1 - \tau_0)/(1 - \tau_0)$ .                                                                     | Proved; checked in <code>scripts/verify_relative_package</code> .                                                |
| R1 weak-vote-passive pure-threshold unanimity | Theorem             | Under the R1 threshold-order domain and weak-vote-passive assessment, admissible pure-threshold candidates are pooling, low-only, and no-information rejection; the selected value maximizes the proposer payoff and uses the H-payoff-minimizing tie-break. | Derived within weak-vote-passive pure-threshold class; checked in <code>scripts/verify_relative_package</code> . |

| Result                               | Type        | Claim                                                                                                                                                                                                                                                                                | Status                                                                                                                                           |
|--------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Majority no-screening                | Proposition | Under $0 \leq a_0^M < a_1^M \leq \bar{y}$ , the no-H, no-screening majority path weakly dominates the considered H-including proposals iff $a_0^M \geq \beta/m$ . If $a_0^M < \beta/m$ , majority can have a screening motive because H is a cheap coalition partner at low beliefs. | Payoff comparison derived with voting-continuation assessment for threshold proposals; checked in <code>scripts/verify_relative_package</code> . |
| Weak-state entry/nesting             | Proposition | In the no-cheap-H majority domain, $V_W^U(\mu) \leq V_W^M(\mu) = 1/m$ for every $\mu$ , so $F_U(\chi) \subseteq F_M(\chi)$ for every $\chi$ .                                                                                                                                        | Proved; checked in <code>scripts/verify_relative_package</code> .                                                                                |
| Conditional H comparison             | Proposition | On $F_U(\chi)$ , H ranks unanimity versus majority by the sign of $\Delta_H(\mu)$ .                                                                                                                                                                                                  | Proved; checked in entry/nesting and classification scripts.                                                                                     |
| Institutional classification         | Corollary   | Under the weak-vote-passive pure-threshold selected value and stated tie-breaking convention, the five categories $\mathcal{C}_0, \mathcal{C}_M, \mathcal{C}_U^H, \mathcal{C}_T^H, \mathcal{C}_M^H$ partition all beliefs and handle entry and H-preference ties explicitly.         | Proved; checked in <code>scripts/verify_relative_package</code> .                                                                                |
| Reduced-surplus majority stress test | Extension   | If weak-only majority surplus is $\rho < 1$ , the clean no-H majority benchmark requires $\rho \geq \bar{\rho}^M = \max\{1 - \tau_0, (1 - a_0^M)/(1 - \beta/m)\}$ .                                                                                                                  | Derived as interpretive stress test; checked in <code>scripts/verify_relative_package</code> .                                                   |

| Result                                     | Type      | Claim                                                                                                                                                                                                                                                                                                  | Status                                                              |
|--------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Initial-recognition<br>agenda-power bounds | Extension | If only the initial<br>Round-1 recognition<br>probability changes to<br>$\pi_H > 0$ while<br>continuations use the<br>$\pi_H = 0$ baseline,<br>expected payoffs<br>decompose into<br>weak-proposer values<br>plus the unresolved<br>H-proposer branch;<br>selection-free bounds<br>bracket the latter. | Bounded; checked in<br><code>scripts/verify_relative_package</code> |
| Working numerical<br>illustration          | Corollary | For $N = 13, \beta =$<br>$0.9, d_0 = 0.19, d_1 =$<br>$0.285, b_0 = b_1 = 0,$<br>classification is given<br>by (illustrative-<br>classification).                                                                                                                                                       | Illustrative and<br>checked.                                        |

### 7.6.7 Result status

General results: terminal R2 unanimity, the selected R1 weak-vote-passive pure-threshold unanimity value, the majority no-screening payoff comparison, weak-state nesting, conditional H comparison, and the complete institutional classification are derived under the fixed-pie relative-package  $\pi_H = 0$  primitives and weak-vote-passive voting assessment.

Sufficient results: the primitive conditions  $0 \leq \tau_0 < \tau_1 \leq \bar{y} \leq 1$ ,  $0 \leq a_0(1) \leq a_1 \leq \bar{y}$ ,  $\beta \leq 1$ , and  $a_0^M \geq \beta/m$  are sufficient for the no-screening majority benchmark, weak-state nesting, and the five-way institutional classification.

Illustrative numerical results: the working example satisfies the sufficient conditions, has a uniform weak-state nesting gap of 0.021375, and yields a single hegemon-preference cutoff at  $\mu = 0.7$  when unanimity forms. It should not be described as an empirical calibration without external discipline for the parameters.

Pending results: global characterization of all standard PBEs remains outside the baseline claim; exact recursive  $\pi_H > 0$  payoffs remain unresolved because continuation values and the H-proposer signaling branch are selection-dependent; the case  $a_0^M < \beta/m$  belongs to a separate majority-screening extension; and alternative proposer tie-breaking rules should be documented before the final paper rewrite.

## 8 Reproducibility Plan

Create new scripts rather than modifying archived proof scripts in place. Stable wrapper scripts matching the redesign agenda:

1. `scripts/verify_relative_package_R2.R`



2. `scripts/verify_relative_package_R1.R`
3. `scripts/verify_relative_package_majority.R`
4. `scripts/verify_relative_package_dominance.R`
5. `scripts/verify_relative_package_entry_nesting.R`
6. `scripts/verify_relative_package_calibration.R`

Detailed baseline and extension scripts called directly or through the wrappers:

1. `scripts/verify_relative_package_R2_piH0.R` (created and run for fixed-pie R2)
2. `scripts/verify_relative_package_R1_piH0.R`
3. `scripts/verify_relative_package_majority_piH0.R`
4. `scripts/verify_relative_package_entry_nesting_piH0.R`
5. `scripts/verify_relative_package_classification_piH0.R`
6. `scripts/verify_relative_package_margins_piH0.R`
7. `scripts/verify_relative_package_robustness_piH0.R`
8. `scripts/verify_relative_package_rho_majority_piH0.R`
9. `scripts/verify_relative_package_piH_bounds.R`

Each result should be independently verified before being promoted in the proof ledger.

## 9 Proof Ledger

| Object                                 | Status                       | Notes                                                                                                                                                                                                                                        |
|----------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Archived feasibility-branch derivation | Archived                     | Preserved in tag <b>redesign-feasibility-branch-2026-05-11</b> ; do not use as current architecture.                                                                                                                                         |
| Relative-package architecture          | Adopted                      | Packages are feasible in every state; screening comes from type-dependent participation thresholds.                                                                                                                                          |
| Minimal H payoff primitive             | Adopted                      | $U_H(y, \theta) = y + b_H(\theta)$ .                                                                                                                                                                                                         |
| Simultaneous public voting protocol    | Adopted for next derivations | The proposer is counted as voting yes; all other states vote simultaneously; individual votes are publicly revealed after the ballot; unanimity uses quota $N$ , majority uses quota $q = \lfloor N/2 \rfloor + 1$ .                         |
| Weak-vote-passive assessment           | Adopted for baseline         | Weak-state deviations do not update beliefs about H because weak states do not observe $\theta$ ; informative histories are disciplined by H's vote. This is a maintained assessment, not a refinement or unrestricted-PBE characterization. |

| Object                                    | Status                                                                       | Notes                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Screening condition                       | Adopted as target primitive                                                  | Need $y_1^*(\mu') > y_0^*(\mu')$ , equivalently $\beta C_H(1, \mu') - b_H(1) > \beta C_H(0, \mu') - b_H(0)$ .                                                                                                                                                                                                                                                               |
| Weak-state cost of concession             | Adopted for fixed-pie reset                                                  | $y$ is costly one-for-one to the weak coalition; accepted weak surplus is $1 - y$ ; weak proposer keeps residual after weak voters receive continuation values.                                                                                                                                                                                                             |
| R2 unanimity under relative packages      | Proved under fixed-pie baseline; checked numerically                         | Weak proposer chooses low-only for $\mu \leq \mu_2^*$ and pooling for $\mu > \mu_2^*$ , where $\mu_2^* = (\tau_1 - \tau_0)/(1 - \tau_0)$ .                                                                                                                                                                                                                                  |
| R1 unanimity under relative packages      | Derived weak-vote-passive pure-threshold selected value; checked numerically | Pooling, low-only, and rejection/continuation are derived from simultaneous public voting plus the maintained weak-vote-passive assessment; the selected value requires proposer argmax and the H-payoff tie-break.                                                                                                                                                         |
| Majority under relative packages          | Derived and checked numerically                                              | Low-only is blocked when $a_0(1) \geq a_1$ . Checked in <code>scripts/verify_relative_package_R1_piH</code> . No screening under $\pi_H = 0$ requires the no-cheap-H condition $a_0^M \geq \beta/m$ ; otherwise H may be a cheap majority coalition partner. Voting-continuation assessment is specified for no-H, pooling-with-H, and low-only-with-H threshold proposals. |
| Reduced-surplus majority/OPEC stress test | Derived and checked numerically                                              | If weak-only majority without H creates surplus $\rho < 1$ , the clean no-H majority benchmark survives only for $\rho \geq \bar{\rho}^M$ . In the working numerical illustration, $\bar{\rho}^M = 0.896216$ . Checked in <code>scripts/verify_relative_package_rho_ma</code> .                                                                                             |

| Object                                                 | Status                                    | Notes                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial-recognition $\pi_H > 0$<br>agenda-power bounds | Bounded and checked<br>numerically        | Expected payoffs are exact at $\pi_H = 0$ and bounded when only the initial Round-1 recognition probability changes to $\pi_H > 0$ , holding continuations at the $\pi_H = 0$ baseline. Checked in <code>scripts/verify_relative_package_piH_bou</code>                   |
| Entry/nesting                                          | Proved and checked                        | Under the weak-vote-passive R1 unanimity value and no-cheap-H majority condition, $V_W^U(\mu) \leq V_W^M(\mu) = 1/m$ for every $\mu$ , so $F_U(\chi) \subseteq F_M(\chi)$ for every entry cost. Checked in <code>scripts/verify_relative_package_entry_r</code>           |
| Conditional institutional<br>comparison                | Derived and checked as sign<br>comparison | For beliefs in $F_U(\chi)$ , H prefers unanimity iff $\Delta_H(\mu) = V_H^U(\mu) - V_H^M(\mu) \geq 0$ . The sign is not globally fixed; the working numerical illustration has a cutoff at $\mu = 0.7$ .                                                                  |
| Institutional classification                           | Proved and checked                        | Five-way classification covers no formation, majority only, both with H preferring unanimity, both with H indifferent, and both with H preferring majority. Entry ties and H-preference ties are explicit. Checked in <code>scripts/verify_relative_package_classi</code> |
| Exact recursive $\pi_H > 0$ game                       | Pending                                   | Bounds are recorded for the initial-recognition stress test, but exact payoffs require rederiving continuation values and solving the H-proposer signaling branch.                                                                                                        |